Lemma. Let $f:[a,b]\to\mathbb{R}$ be a continuous function s.t. differentiable on (a,b). If $f(a)=f(b)$, then there exists $c\in(a,b)$ such that $f'(c)=0$.

Proof. If f is constant, then there is nothing to prove. Assume f is not constant. Suppose $f(x)>f(a)$ for some $x\in(a,b)$. Then, by the min-max Theorem, there exists $c\in(a,b)$ such that $f(c)\geq f(x), \forall x\in[a,b]$. Thus, $f'(c)=0$. ∎

Mean-value Theorem.

Let $f:[a,b]\to\mathbb{R}$ be a continuous function s.t. differentiable on (a,b). Then, there exists $c\in[a,b]$ such that $f(b)-f(a)=f'(c)(b-a)$.

Proof. Take $g(x)=f(x)-([[f(b)-f(a))/(b-a)]\cdot(x-a)+f(a))$. Then, $g(a)=g(b)=0$, so $g'(c)=0$ for some $c\in[a,b]$. ∎

Generalized Mean-Value Theorem.

Let $f, g:[a,b]\to\mathbb{R}$ be a continuous function s.t. differentiable on (a,b). Then, there exists $c\in[a,b]$ such that

$$g'(c)[f(b)-f(a)]=f'(c)\cdot[g(b)-g(a)].$$

Proof. Take $h(x)=g(x)[f(b)-f(a)]-f(x)[g(b)-g(a)]$.

Then,
$$h(a)=g(a)f(b)-g(a)f(a)-f(a)g(b)+f(a)g(a)$$
$$h(b)=g(b)f(b)-g(b)f(a)-f(b)g(b)+f(b)g(a).$$ ∎

L'Hôpital's Theorem.

Let f, g be differentiable on $B_r(x_0) \subseteq \mathbb{R}$, for some $r>0$ and x_0.

If $\lim_{x \to x_0} f(x) = \lim_{x \to x_0} g(x) = 0$ and $\lim_{x \to x_0} \frac{f'(x)}{g'(x)}$ exists,

then $$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)}.$$

Proof. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |t - x_0| < \delta \Rightarrow \left| \frac{f'(t)}{g'(t)} - \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \right| < \varepsilon$$

Fix such t, $\frac{f(t)}{g(t)} = \frac{f(t) - 0}{g(t) - 0} = \frac{f(t) - \lim_{x \to x_0} f(x)}{g(t) - \lim_{x \to x_0} g(x)} = \frac{f'(c)}{g'(c)}$ for some $0 < |c - x_0| < \delta$

Now, $\left| \frac{f(t)}{g(t)} - \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \right| = \left| \frac{f'(c)}{g'(c)} - \lim_{x \to x_0} \frac{f'(x)}{g'(x)} \right| < \varepsilon$ for any $0 < |t - x_0| < \delta$,

So Proved. ☒

「Tayler's Theorem.

Let $f:[a,b]\to\mathbb{R}$ be given and $n\in\mathbb{N}$ be fixed.

If $f^{(n)}$ exists with Continuous on $[a,b]$ and differentiable on (a,b).

Then, given $x\in[a,b]$, there exists $C_x\in(a,x)$ such that

$$f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}\cdot(x-a)^k+\frac{f^{(n+1)}(C_x)}{(n+1)!}\cdot(x-a)^{n+1}$$

Proof. Given $x\in[a,b]$, define $F(t)=\sum_{k=0}^{n}\frac{f^{(k)}(t)}{k!}\cdot(x-t)^k$

and for $g(t)=(t-a)^{n+1}$, $G(t)=\sum_{k=0}^{n}\frac{g^{(k)}(t)}{k!}\cdot(x-t)^k$.

Then,
$$F'(t)=\frac{d}{dt}\left(\sum_{k=0}^{n}\frac{f^{(k)}(t)}{k!}(x-t)^k\right)$$
$$=\sum_{k=0}^{n}\frac{d}{dt}\left(\frac{f^{(k)}(t)}{k!}\cdot(x-t)^k\right)\qquad\left(\frac{f^{(k+1)}(t)}{k!}\cdot(x-t)^k-\frac{f^{(k)}(t)}{(k-1)!}\cdot(x-t)^{k-1}\right)$$
$$=\frac{f^{(n+1)}(t)}{n!}\cdot(x-t)^n\qquad(\text{Telescoping})$$

And $G'(t)=(n+1)\cdot(x-t)^n$ $\left(g'(t)=(n+1)(t-a)^n,\ g^{(2)}(t)=(n+1)\cdot n\cdot(t-a)^{n-1}\cdots,\ g^{(n+1)}(t)=(n+1)!\right)$

Now, by the generalized Mean-Value Theorem,

$$\underbrace{F'(C_x)\cdot[G(x)-G(a)]}_{\frac{f^{(n+1)}(C_x)}{n!}\cdot(x-C_x)^n\cdot\left[(x-a)^{n+1}-0\right]}=\underbrace{G'(C_x)\cdot[F(x)-F(a)]}_{(n+1)\cdot(x-C_x)^n\cdot\left[f(x)-\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}\cdot(x-a)^k\right]}$$

So, Proved.

Some Application of Taylor.

Prop. Let $f : (a,\infty) \to \mathbb{R}$ be a twice-differentiable function and

$$M_i \overset{\text{def}}{=} \sup_{x>a} |f^{(i)}(x)| \text{ exists.}$$

Then, $M_1^2 \leq 4 \cdot M_0 \cdot M_2$. Moreover, if $f(x) \to 0$ as $x \to \infty$, then $f'(x) \to 0$ as $x \to \infty$

Proof. Let $x \in (a,\infty)$ be fixed. Given $h>0$, the Talyor's Theorem gives

$$f(x+2h) = f(x) + f'(x)\cdot 2h + \frac{f''(c_x)}{2!}\cdot(2h)^2 \text{ for some } c_x \in (x, x+2h).$$

So, $f'(x) = \frac{1}{2h}\cdot\left(f(x+2h) - f(x)\right) - h\cdot f''(c_x)$, so

$$|f'(x)| = \frac{1}{2h}\cdot\left[|f(x+2h)| + |f(x)|\right] + h\cdot|f''(c_x)| \leq \frac{1}{2h}\cdot 2\cdot M_0 + h\cdot M_2 = \frac{M_0}{h} + hM_2.$$

Now, take $h = \sqrt{M_0/M_2}$, then $\frac{M_0}{h} + hM_2 = \sqrt{M_0}\cdot\sqrt{M_2} + \sqrt{M_0}\sqrt{M_2} = 2\cdot\sqrt{M_0 M_2}$.

(If $M_2 = 0$, the f' is Constant, so f is $f'(0)\cdot x + b$, $f'(0)$ must be zero.).

The second assertion is proved by;

$$\sup_{x>a}|f'(x)| \leq \sup_{x>a}|f''(x)| \cdot \sup_{x>a}|f(x)| \to 0 \text{ as } a \to \infty.$$